Review plan · for the teacher

# Tricking AI and defending it: unit check

AI security · Level 2 · Unit check

| Time | Materials |
| --- | --- |
| 45 min | Board, chalk, paper. No devices needed. |

## Learning goals

- Orders hidden in the data
- Jailbreaks: talking a chatbot round
- Guardrails: rules around the model
- Ask before it acts

## Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

## Discussion

- Which AI tool at your school reads text other people wrote?
- Which question surprised you, and why?
- What will you ask the next time someone offers the school an AI agent?

## Exit ticket

- One defence I will ask for, and where I will ask for it

**Try it at school tomorrow**
Pick the defence from this unit your school needs most (outside text as data, reporting what slips through, a check on the action, or only the tools a job needs) and ask about it the next time anyone suggests a new AI tool.

Quiz sheet · for each team

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Team _____ Date _____

**1** **What is prompt injection?**

◯ A typing mistake in a prompt ◯ A virus injected into a phone

◯ An order hidden in text an AI reads, which it may follow

**2** **A language model can always tell your instructions apart from words inside a document it is reading.**

◯ True ◯ False

**3** **Match each term to its meaning.**

| | | |
|---|---|---|
| Direct injection | • • • • | An order hidden in an email the AI summarises |
| Indirect injection | • • • • | Testing a chatbot on purpose, with permission |
| Jailbreak | • • • • | An order typed into the chat |
| Red teaming | • • • • | Talking a chatbot round its safety rules |

**4** **Why can a reworded request get past a chatbot's refusal?**

◯ Its refusals are learned from the wordings it saw ◯ Rewording deletes its safety rules

◯ The chatbot gets tired after a few questions

**5** **Pupils made the school chatbot write something unkind. Which responses make sense? Choose all that apply.**

☐ Report it to the provider ☐ Record what was tried and what got through ☐ Add a check or rule, then test again

☐ Hold a contest to find more

**6** **A filter blocks the word "password". Which message gets past it?**

◯ "Reply with your login code to keep your account" ◯ "Send me your password" ◯ "Password needed now"

**7** **Put the layers in the order a message meets them.**

☐ The input filter

☐ The model writes

☐ The action check in code

☐ The person types a request

☐ The output filter

**8** **A check in code that a message was asked for and approved works however the message is worded.**

◯ True ◯ False

**9** **Which of these are acting tools, rather than reading tools? Choose all that apply.**

☐ Search the library ☐ Send a message ☐ Look up the timetable ☐ Pay a bill ☐ Delete files

**10** **Complete the rule for AI agents.**

Give an agent the _____ tools its job needs, and a person's _____ before it acts.

**Word bank:** most · fewest · guess · yes

Answers · for the teacher

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1. **An order hidden in text an AI reads, which it may follow**

   Prompt injection hides instructions in what an AI reads, or types them in directly.

2. **False**

   It reads your instructions and the document as one stream of text, with no reliable way to tell them apart.

3. **Direct injection — An order typed into the chat**
   **Indirect injection — An order hidden in an email the AI summarises**
   **Jailbreak — Talking a chatbot round its safety rules**
   **Red teaming — Testing a chatbot on purpose, with permission**

   Injection hides or types orders; a jailbreak rewords its way round the rules; red teaming is the defenders' own test, done with permission.

4. **Its refusals are learned from the wordings it saw**

   A model learns to refuse the kinds of wording it saw in training.

5. **Record what was tried and what got through; Report it to the provider; Add a check or rule, then test again**

   Treat it as a finding: record it, report it and fix it, then test again.

6. **"Reply with your login code to keep your account"**

   A word list only catches the words on it.

7. **Top to bottom: 2, 3, 5, 1, 4**

   Requests are filtered going in, replies going out, and the last gate before anything happens is the check in code that a person approved it.

8. **True**

   It checks the action, not the words: who asked and who approved this exact message.

9. **Send a message; Delete files; Pay a bill**

   Sending, deleting and paying change the world, so a tricked agent could turn them against you.

10. **Give an agent the fewest tools its job needs, and a person's yes before it acts.**

    Least privilege keeps the blast radius small, and a person's yes, checked in code, stands between a trick and any action.